Risk Reasons PRD

Problem & Opportunity

When a shipment goes Medium or High Risk, customers see the flag but not the cause. A carrier breakdown, a facility holding the truck, a GPS signal that stopped updating, and a delivery window that's simply tight all look the same, and each calls for a different response. So customers contact a CET, who pieces together the answer from ETA context, tracking freshness, check calls, and carrier updates and explains it by hand. Tracking costs Operations about $6M a year, and a meaningful share of that effort is this repeated explanation.

Show one supported reason on each at-risk load, chosen by rules we can audit. If customers can see why a load is at risk, fewer of them need to ask, and the CETs who still get asked can answer faster. Our goal is $450K in annual savings, about 7.5% of tracking cost. We've already run the logic in shadow mode, where it covered 70% of Medium and High Risk loads, and measured the change in outreach tickets and in Ops time per ticket. The next step is taking it live.

Target User

Customers monitoring in-flight loads, and the CETs who answer their questions. The customer's job: *"When my shipment is flagged at risk, tell me what's causing it, so I know whether to act or wait."* The CET's job is the mirror image: answer fewer repeat questions and spend that time on loads that need intervention.

Solution - End to End

1. A load's Risk Level is Medium or High, which is the only time reason selection runs.
2. The logic checks each reason's trigger in priority order: carrier-reported, then facility-related, then tracking visibility, then ETA/model.
3. The highest-priority eligible reason becomes the single active reason. It's returned with the stop it affects and the time it occurred. Every other eligible reason is stored as passive.
4. The logic re-runs every 30 minutes. If a newer eligible reason appears at least four hours after the active one, it takes over and the old one becomes passive. Tight Delivery Window never triggers this swap.
5. It stops when the load goes Low Risk, Delivered, or Cancelled.

Product Concepts

- **Active and passive reasons:** Active means the reason currently explains the displayed risk level. Passive means it happened and may still help as context, but no longer explains the current risk: Only one reason is active at a time.
- **Priority by specificity:** Specific, attributable causes outrank general ones. If a carrier reports a late ETA and tracking is also limited, the carrier reason wins.

- **Affected stop:** Each reason is tied to a stop, so the explanation says where the problem is, not just that one exists.
- **Reason versus risk score:** Reasons are rule-triggered and separate from the ML Risk Level. A displayed reason means a supported cause is present, not that it drove the score.

Priority	Category	P1 reasons
1	Carrier-reported	Mechanical issue, accident, carrier-communicated late ETA
2	Facility-related	Carrier in detention, had detention, pretendered load
3	Tracking visibility	Limited tracking visibility, GPS not moving
4	ETA/model	Tight delivery window

Tradeoffs

- **Priority first, recency after four hours:** Priority keeps the answer stable and the override stops a stale explanation from sticking around. The cost is that a lower-priority reason can push out a higher-priority one that's still true. The four-hour window is a guess until we test it.
- **One reason, not several:** A single answer is easier to act on than a list. Real causes are often compound, but stored passive reasons leave room for P1.
- **Out of P1:** weather, carrier and facility history, customer preferences, and appointment changes. Each needs validation before customers see it.

Metrics

- **Topline:** Outreach tickets per 100 Medium and High Risk loads, by topic, against a holdout. CETs tag each ticket as delay or risk reason, ETA change, carrier update, status confirmation, or other. Success is a drop in reason-related topics with no rise in the others.
- **Ops efficiency:** Average Ops time per ticket, and dollar savings against the $450K goal.
- **Quality:** Audited accuracy by category, with proposed gates of 90% for carrier and tracking reasons and 85% for facility reasons. Coverage by category, with Tight Delivery Window reported separately, since it explains the least.
- **Guardrails:** How often the active reason changes per load, and whether tickets in non-reason topics rise.

Testing and implementation

Shadow mode is done. Before going live, we backtest on past Medium and High Risk loads where CETs recorded the cause, and check how often our reason matches theirs. We also use the backtest to test our design choices: the override window (2, 4, 6, and 8 hours, picking the shortest that avoids flip-flopping without leaving stale reasons up) and the priority order against the cause CETs named as primary.

The live rollout goes to a limited set of customers, with the holdout assigned by customer rather than by shipment, since customers raise all their loads with the same CETs. We expand when accuracy meets the gates, savings track toward the goal, and non-reason tickets don't rise.

Two spec gaps need closing first. We need a rule for an active reason whose trigger clears while the load stays Medium or High Risk, and an owner for marking the active reason passive when the load goes Low, Delivered, or Cancelled.

Next steps and vision

Next: approve the limited live rollout and the backtest, and have Data Science measure how often displayed reasons agree with the risk score.

Then: add appointment changes with light default weight, since many are customer-requested, and use the stored passive reasons to show more than one cause where it helps. Weather and carrier and facility history follow once Data Science validates their predictive value.

Longer term: Data Science decides whether these signals stay an explanation layer on top of Risk Level or feed a new parent risk model. If reasons prove accurate and trusted, they could also drive proactive updates when the reason changes, so customers hear about a problem before they ask.